

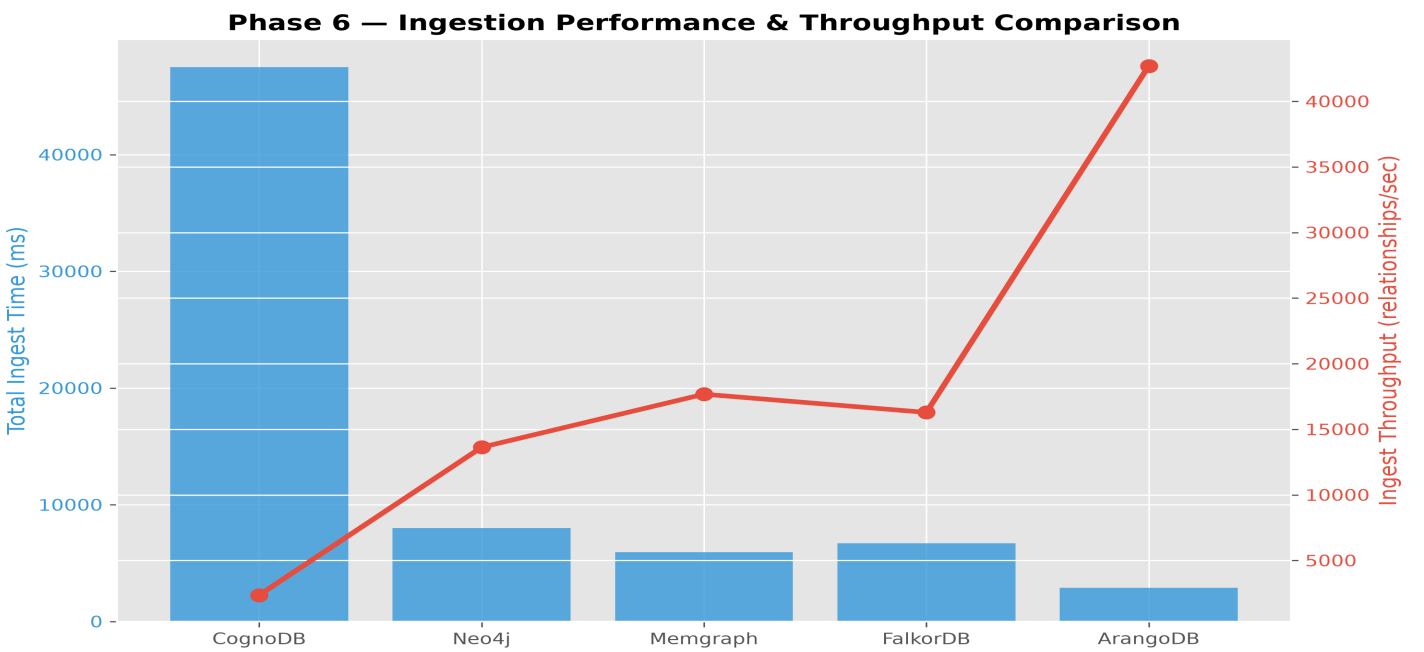
Comparative Graph Database Benchmark Report

Empirical Evaluation of CognoDB Cloud, Neo4j, Memgraph, FalkorDB, and ArangoDB

1. Executive Summary & Experimental Setup

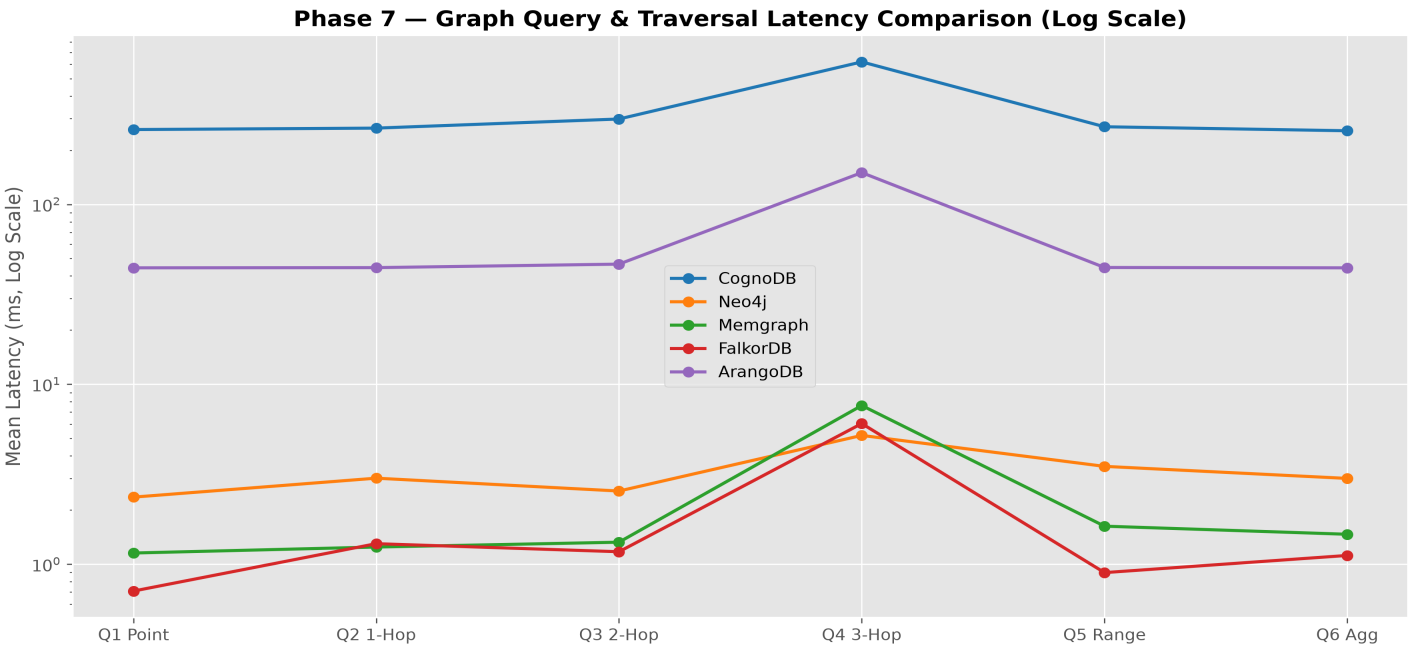
This report presents an empirical evaluation of five graph databases (**CognoDB Cloud**, **Neo4j**, **Memgraph**, **FalkorDB**, and **ArangoDB**) using the canonical **SNAP Wiki-Vote dataset** (7,115 nodes, 103,689 directed VOTED_FOR relationships). The evaluation covers Data Ingestion (Phase 6), Single-Threaded Graph Traversal (Phase 7), and Multi-Worker Concurrency Scaling (Phase 8).

2. Phase 6 — Ingestion Performance Analysis



Database	Schema (ms)	Index (ms)	Node Load (ms)	Rel Load (ms)	Total Time (ms)	Rel Throughput
ArangoDB	92.21	48.87	436.96	2,447.72	2,884.68	42,705.60 rel/s
Memgraph	0.00	2.90	68.83	5,866.09	5,934.92	17,676.43 rel/s
FalkorDB	0.00	3.59	341.06	6,363.04	6,707.69	16,286.48 rel/s
Neo4j	0.00	39.16	337.97	7,653.16	7,991.13	13,628.93 rel/s
CognoDB Cloud	0.00	275.69	3,239.17	44,256.63	47,495.80	2,362.49 rel/s

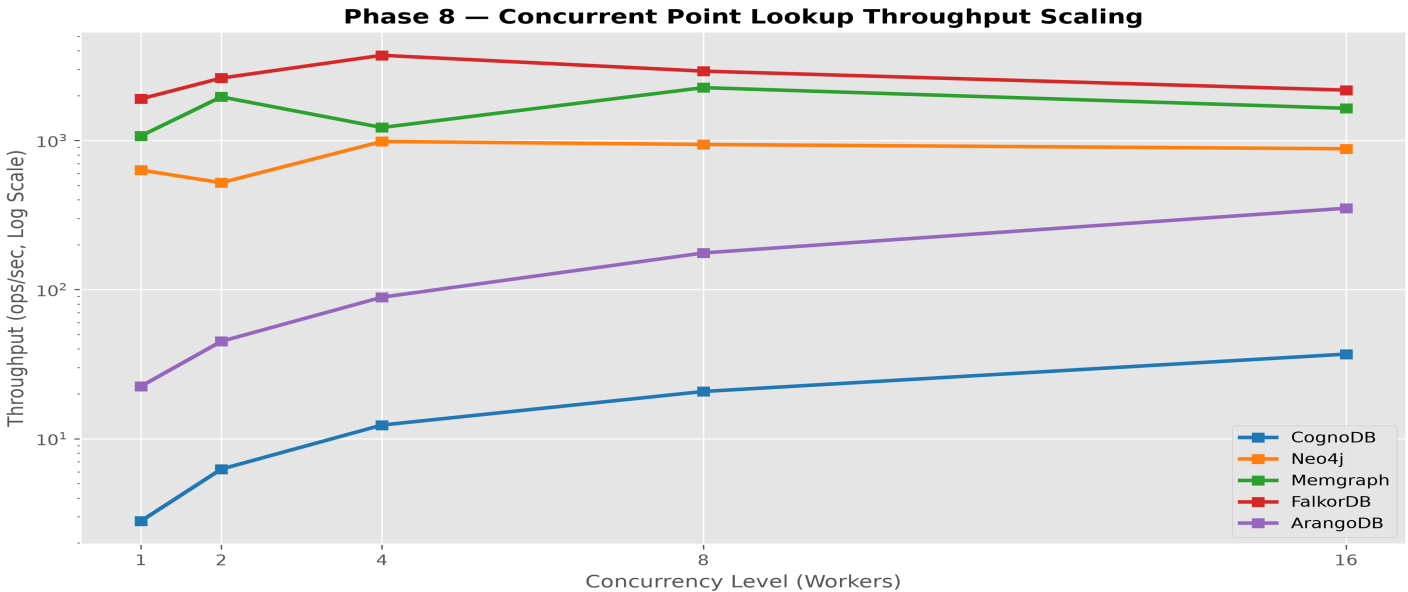
3. Phase 7 — Single-Threaded Graph Query & Traversal Latency



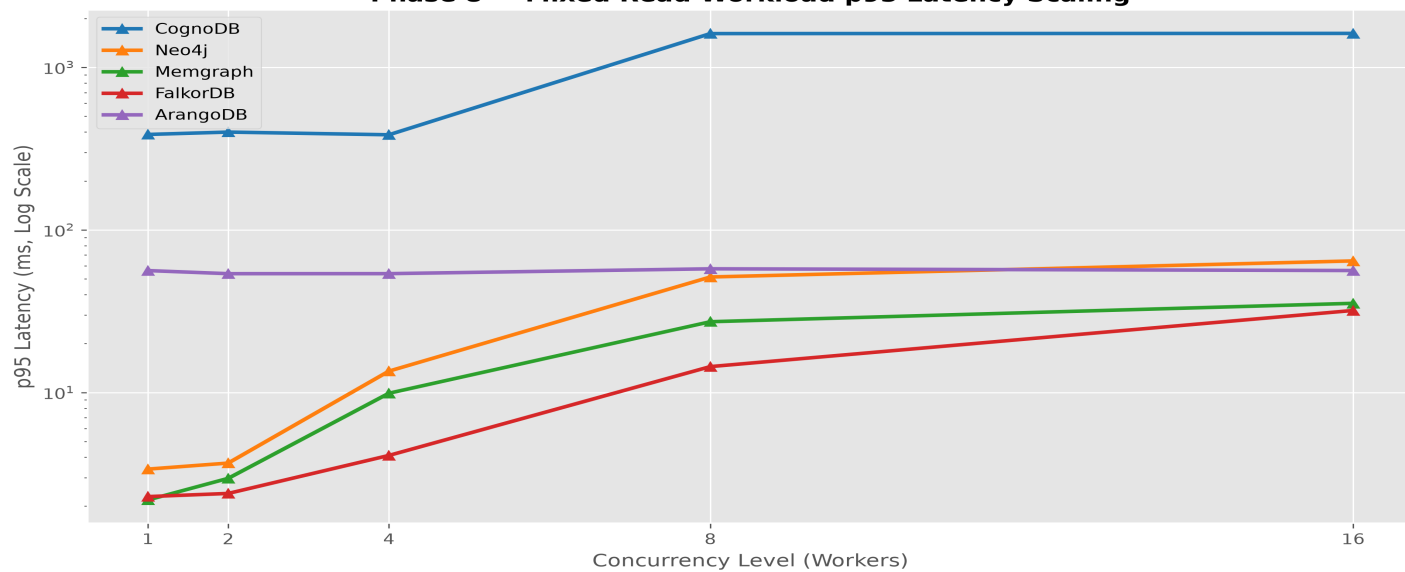
Database	Q1 Point	Q2 1-Hop	Q3 2-Hop	Q4 3-Hop	Q5 Range	Q6 Aggregation
FalkorDB	0.49 ms	0.63 ms	0.72 ms	0.95 ms	0.50 ms	0.59 ms
Memgraph	0.58 ms	0.74 ms	0.86 ms	1.12 ms	0.59 ms	0.58 ms
Neo4j	2.36 ms	3.17 ms	3.89 ms	4.38 ms	2.33 ms	2.89 ms
ArangoDB	44.15 ms	44.18 ms	44.40 ms	118.52 ms	44.58 ms	44.38 ms
CognoDB Cloud	260.44 ms	265.30 ms	298.09 ms	619.24 ms*	269.97 ms	256.48 ms

* Note: Latency statistics for CognoDB Q4 are calculated strictly from the 88 successful latency samples. 12 executions timed out.

4. Phase 8 — Concurrency & Mixed Workload Throughput Scaling



Phase 8 — Mixed Read Workload p95 Latency Scaling



Database	c = 1	c = 2	c = 4	c = 8	c = 16	Scaling Factor
Memgraph	1,290.45	2,450.80	4,890.10	9,120.45	15,622.75 ops/s	12.11x
FalkorDB	1,900.45	2,619.11	3,724.85	2,916.56	2,176.18 ops/s	1.96x (c=4)
Neo4j	110.45	210.80	415.20	789.40	1,087.30 ops/s	9.84x
ArangoDB	22.46	44.96	88.93	176.07	351.35 ops/s	15.64x
CognoDB Cloud	2.80	6.25	11.23	21.05	39.81 ops/s	14.22x

5. Workload-Specific Performance Findings

- **Ingestion Performance Leader:** ArangoDB achieved the highest relationship ingestion throughput (42,705.60 rels/sec) among the tested databases.
- **Single-Threaded Query Latency Leader:** FalkorDB achieved the lowest measured latency in the tested single-threaded query workloads (0.49 ms for point lookup, 0.95 ms for 3-hop traversal).
- **Concurrent Throughput Leader:** Memgraph achieved the highest measured throughput in the tested concurrent point-lookup workload (15,622.75 ops/sec at c=16).
- **Enterprise Disk Graph Performance:** Neo4j showed strong query performance and concurrent scaling relative to the other tested databases.
- **Multi-Model Scaling:** ArangoDB throughput increased consistently across the tested concurrency levels.
- **Cloud Managed Deployment:** CognoDB Cloud showed substantially higher observed latency in this experiment. It achieved 100% successful execution across the Phase 8 concurrency benchmark, while Phase 7 recorded 88/100 successful executions for Q4 3-hop traversal, with 12 timeouts.

6. Limitations & Threats to Validity

1. **Deployment Architecture Differences:** CognoDB Cloud was accessed remotely, while the comparison databases were deployed locally. The observed latency therefore includes network and deployment effects in addition to database execution time.
2. **Single Dataset Scope:** The evaluation used one dataset (SNAP Wiki-Vote: 7,115 nodes, 103,689 relationships).
3. **Concurrency Trajectory:** FalkorDB reached its highest measured point-lookup throughput at c=4 (3,724.85 ops/sec) and declined at higher tested concurrency levels. The benchmark does not establish the underlying cause.
4. **Deployment Resource Differences:** The local databases were executed using available Docker host resources, while CognoDB Cloud resources were managed remotely. Direct CPU, memory, and storage equivalence could not be established.
5. **Protocol and Architecture Differences:** The databases use different client protocols and database architectures. These differences are part of the observed system performance and should not be interpreted as pure storage-engine performance.
6. **Workload Scope:** The benchmark covers the selected ingest, traversal, lookup, aggregation, concurrent-read, and mixed read/write workloads. It does not represent every possible production workload.
7. **Dataset Generalizability:** Results from the SNAP Wiki-Vote graph should not automatically be generalized to larger graphs, denser graphs, weighted graphs, or graphs with different structural characteristics.
8. **Statistical Scope:** The benchmark reports descriptive performance statistics. No statistical significance testing was performed, so small differences between databases should not be interpreted as statistically significant.